Data Structures

collection of objects

heterogeneous

- **lists** (arrays)
  - → ordered
  - ↔ mutable
- **tuples** (special lists)
  - → immutable
- **sets**
  - unique collection of objects
- **Dictionaries** (hash maps)

Slicing

a[1] / a[-6]

left to right (+ve)

a = [10 , 20 , 30 , 40 , 50 , 60 , 70]

right to left (-ve)

a[5] / a[-2]

a = 45 + 54

evaluation & assignment

a , b = b , a

gets packed into a tuple

(b , a)

get unpacked into variables

sets → unique

set(4, 5, 6, 5, 7, 7, 7, 8, 10, 10)

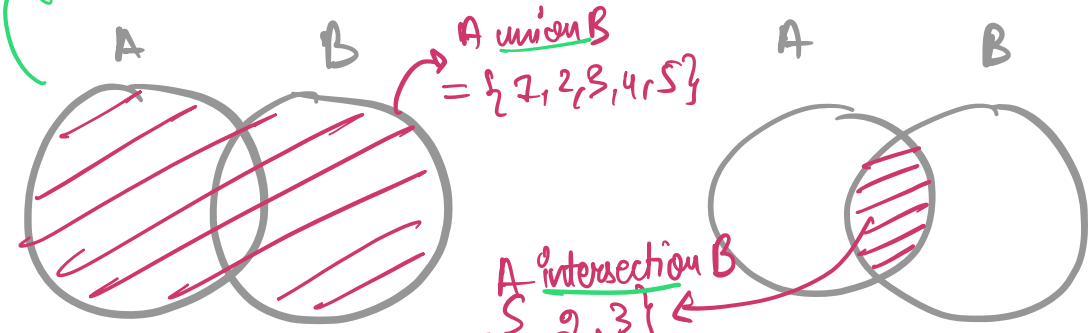
↓

unique values only → {4, 5, 6, 8, 7, 10}

venn diagram

$A = \{1, 2, 3, 4\}$

$B = \{2, 3, 5\}$



A union B $= \{1, 2, 3, 4, 5\}$

A intersection B $= \{2, 3\}$

A − B (difference)

(A − (A intersection B))

$= \{1, 4\}$

(A − B) union (B − A)

(A union B) − (A intersection B)

$= \{1, 4, 5\}$

symmetric difference